

Dynamic Connectedness between commodities, forex exchanges, equity markets, and uncertainty: Evidence from the US and Australia

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Abstract

The financial markets are facing a situation of growing unprecedented uncertainty, supported by various shocks to the system, such as pandemics or energy and inflation pressures. This paper studies the interdependence across extreme quantiles of volatility of related financial markets, such as stocks, foreign exchange, oil, and gold, under the impacts of market sentiment. Quantile Time-Varying Parameter Vector Autoregression (TVP-VAR) is employed by the project as a new approach that is superior to the traditional mean-based measures. This model performs well in extreme market conditions to check for dynamic spillovers related to return and energy shocks. The empirical results could show evidence of the density of network connections in not only the mean (median) cases but also the lower and upper extremes of market volatility. In addition, time-varying effects analysis will enable the identification of significant shock events, which include the Shale Oil Crisis, the COVID-19 pandemic, and recent energy shocks related to the Russia-Ukraine conflict. These findings will be critical for investors, financial markets, fund managers, and policymakers regarding the management of risks, especially at times when there are significant disruptions in the market

Keywords: *TVP-VAR, uncertainty, interdependence, extreme market conditions, network connections, significant disruptions, extremes of market volatility, shock events*

1. Introduction

In the last decades, the growing interconnectedness of financial and economic variables, including those of commodity prices, stock markets, and macroeconomic indicators, makes knowledge about systemic risk necessary. Global crises like the 2008 financial crisis, COVID-19, and geopolitical tensions like the Ukraine-Russia conflict, increase market volatility and risk levels, which lead to the need to investigate these relations via the proxies of market uncertainty sentiment indices. While several pieces of literature have researched conventional financial assets, such as the equity market and forex (Barunik et al., 2017; Chow, 2021; Wang et al, 2023), those papers might not include the effects of commodities on the whole system. For instance, the two most important commodities that strictly influence the financial and macroeconomic systems are crude oil and gold. Their nature of characteristics, for instance, oil is used in production and energy markets, and gold is acting as a safe-haven asset (Baur and McDermott, 2010). These are very critical to understanding how the shocks spread across the economy, especially in the US and Australia as comparison case studies.

However, conventional methodologies of risk assessment, such as correlation analysis or traditional VaR, usually focus on average market conditions while missing conducting necessary studies on the varying dynamics that occur under different quantile levels, such as extreme market conditions. Thus, there is a need for a more appropriate and accurate approach to examine how these economic variables interact under normal and extreme conditions.

2. Theoretical Framework

Chisti et al. (2) researched an empirical inquiry into how various economic indicators, specifically foreign institutional investments, exchange rates, and crude oil prices, are interrelated and impact stock market performance in relation to the Nifty 50 index in the Indian financial system [10]. Neely researched the impact of the Russia–Ukraine war on the prices of oil and other sanctioned commodities, as did Qi Zhang et al. They all come to the conclusion that when such a big exporter goes offline either because of sanctions or because of infrastructural problems, this inevitably affects the supply of oil [1],[2]. Khan et. al (4) go one step further and analyze how extreme geopolitical events affect freight prices, which are a very important part of pricing commodities [19]. Hung (60) combined wavelet coherence analysis with a multivariate DCC-GARCH model to analyze the time-frequency dynamics between exchange rates, stock markets, and international commodities markets, in particular, gold and oil [12]. Reddy et al. (599) analyzed six macroeconomic indicators, namely inflation, interest rates, gold, silver, crude oil, and exchange rates, for their effects on sectoral performance and pointed out the complex interrelationship among various dimensions of macroeconomic factors and the outcomes in sectors. Rehman and Vo (3) examined the returns integration between commodities in precious metals, energy, and industrial metals, in an attempt to reveal the interconnectivity and interdependencies among these categories of commodities [15]. Shah et al. (2) conducted an in-depth study into the dependence between crude oil, precious metals (including gold), and foreign exchange markets using both time and frequency domains [14]. Finally, making a more solid empirical analysis, Tiwari et al. (1) evaluated the systemic risk and interdependencies across stock market indicators with the oil market of G7 economies [13]. Lastly, Wei et al. (24) combine wavelet techniques with statistics of long memory to explore long-term correlations among Chinese stock market indices and crude oil futures, looking at the effects of more recent financial crises [17]. Fu et al. (2) use wavelet analysis and Non-linear distributed lag modeling (NARDL) to study the relationship between subjective factors, consumer sentiment and inflation expectations, and energy prices, focusing on Brent crude and natural gas. Their study revealed that there is a negative relationship between consumer sentiment and energy prices and a positive relationship between inflation expectations and energy prices [20]. Shang and Hamori (1) analyzed the relationship and spillover effect between crude oil, gold, financial markets and macroeconomic indicators using a quantile time-frequency connectedness approach. Kalman filter was used within the Time-Varying Parameter Vector Autoregression (TVP-VAR) model to estimate the time-varying parameters and later apply the model for forecasting purpose suggestions [21].

In recent years, quantile connectedness has become the focus of academic inquiry. This is due to the ability to detect specific associations in different quantile tails. and severe market conditions. This method is extremely valuable for analyzing financial markets and a wide range of indicators. Such detailed analysis enhances risk assessment and facilitates,

as well as supports, the development of bespoke risk management strategies tailored to different market stress levels (Ando and Bai 270) [3]. Quantile connectedness, such as the Quantile TVP-VAR analysis, not only helps investors detect early signs of market downturns and upswings. But it also serves as an advanced forecasting tool for anticipating market fluctuations and major economic changes. This novel approach helps protect portfolios and capitalize on emerging opportunities. In the same manner, policymakers can also leverage this approach to design responsible policies that adapt to fluctuating market conditions (Ando et al. 2402). This will increase the ability of investors, central banks and government agencies to make informed and flexible decisions. which promotes economic stability... Moreover, quantile connectedness has proven effective in assessing rare but high-impact tail risks. It provides a comprehensive approach that facilitates dynamic analysis of the market under various situations. The quantitative time-frequency method pioneered by Chatziantoniou et al. (2) shows improved integration of time-frequency quantile exposure methods [9]. This innovative framework was developed by Chatziantoniou et al. (4) and Ando et al. (1) and offered a unique ability to examine relationships within the time-frequency spectrum and across quantile ranges. Doing so, the method provides a comprehensive analysis of the interaction between financial markets and indicators under various conditions, as well as helps to gain a deeper understanding of market changes.

3. Methodology

In this study, we utilize the quantile framework of connections initially designed by Chatziantoniou et al. (4) to scrutinize how the effects are transmitted among the commodity, stock, and foreign exchange at various levels of quantiles [8]. Time-varying parameter Vector Autoregressive (TVP-VAR) models are frequently used in econometrics to research the changing relations between numerous sequential data sets, where variables available vary over the years. The concept of ‘linkages’ in this scenario basically refers to how two or more variables or even financial assets possibly relate to each other as time passes and has often been used as an important concern in the context of systemic risk in finance.

This approach is also referred to as the time-frequency connectedness in quantiles, and it enhances the framework of connectedness hence proposed by Diebold and Yilmaz (60), Baruník and Křehlík (41) and Ando et al. (3) [22][6][3]. To put it into the context further, Diebold and Yilmaz (60) laid the foundation of the methodology centered on the multivariate time series analysis, which employs generalized variance decomposition, which overcomes these limitations [22]. Antonakakis et al. (5) then developed a relevant further method using time-dependent parameter vector autoregressive (TVP-VAR) models within the broader theoretical framework of Koop and Korobilis (105) that combines structural and Bayesian techniques [23][24].

In advancing this conception, Baruník and Křehlík (2018) devised means to probe connectedness over various frequency bands and across time, especially in terms of dissecting short-term (high-frequency) and long-term (low-frequency) connectedness. Ando et al. (3) unlocked themselves from the above challenges by coming up with the quantile connectedness method – a method that measures the association/dependency between variables at different levels of quantile, thus, a detailed analysis of extreme events and their impact on the interconnectedness is possible [3].

The quantile connectedness analysis is highly useful in assessing the extent of market connectedness during all forms of market regimes: norm (average market activity), high quantile (bullish market), and low quantile (bearish market activity). This assists in estimating how strong the impacts of some in terms of magnitude and direction of market

shocks or price jump events. Such an approach allows one to better understand how market overtone transformations operate under the influence of crisis situations.

3.1. Model

TVP-VAR model for time series variables at time t can be written as

$$Y_t = \Phi_t Y_{t-1} + \varepsilon_t$$

Where

- Y_t : $K \times 1$ vector of endogenous variables
- Φ_t : $K \times K$ time-varying coefficient matrix changing over time
- ε_t : $K \times 1$ vector of error terms, assumed to follow a multivariate normal distribution with mean = 0 and time-varying covariance matrix Σ_t

3.2. Time-Varying Parameter Evolution

The time-varying coefficient matrix Φ_t follows a stochastic process, which is modelled as a random walk:

$$\Phi_t = \Phi_{t-1} + \eta_t$$

Where

- η_t : $K \times K$ matrix of innovations associated with the time-varying parameters, which is assumed to be normally distributed

3.3. Variance Decomposition (Forecast Error Variance Decomposition - FEVD)

$$\theta_{ij}^H(t) = \frac{\sigma_{ii}^{-1} \sum_{h=0}^{H-1} (e_i' A_h(t) \Sigma_t e_j)^2}{\sum_{h=0}^{H-1} (e_i' A_h(t) \Sigma_t A_h'(t) e_i)}$$

Where

- $\theta_{ij}^H(t)$: portion of the H-step ahead forecast error variance of variable i attributable to shocks of variable j at time t
- $A_h(t)$: time-varying coefficient matrix for the h -th lag in the moving average representation of TVP-VAR
- Σ_t : time-varying covariance matrix of the innovations ε_t
- e_i and e_j are the selection vectors of the i -th and j -th variables
- H : forecast horizon

3.4. Connectedness Measures

3.4.1 Total Connectedness Index (TCI)

TCI measures the overall degree of connectedness in the system. Total connectedness at time t , which represents the average percentage of forecast error variance due to shock from other variables of the model, is defined as:

$$TCI_t = \frac{1}{K} \sum_{i=1}^K \sum_{j=1, i \neq j}^K \theta_{ij}^H(t)$$

3.4.2 Directional Connectedness (from and to others)

The connectedness from variable i to others can be calculated as

$$C_{i \rightarrow}(t) = \sum_{j=1, j \neq i}^K \theta_{ji}^H(t)$$

While the connectedness to the variable i from others can be calculated as:

$$C_{i \leftarrow}(t) = \sum_{j=1, j \neq i}^K \theta_{ij}^H(t)$$

3.4.3 Net Pairwise Connectedness

The net connectedness between two variables i and j can be calculated as:

$$C_{ij}^H(t) = \theta_{ij}^H(t) - \theta_{ji}^H(t)$$

3.5. Dynamic Connectedness Measures

In the TVP-VAR framework, connectedness measures change over time due to the fundamental interconnectivity of the associated variables. This dynamicity allows the examination of such correlations as systemic risk, contagion and co-movements within the financial markets.

The equations provided in previous sections can be used to analyze the time-varying connectedness in the context of systemic risk, macroeconomic policy and financial markets. It is the nature of what happens to the transmitted shock wave as deviation is induced by comparison model sizes.

4. Data

The data in this paper were collected from publicly available online sources, including Federal Reserve Economic Data, investing.com and policyuncertainty.com. The variables used in this research are monthly prices for WTI crude oil, gold, S&P500, USD index, US uncertainty market sentiment, oil uncertainty market sentiment, Brent crude oil, gold, ASX50, AUD index, AU uncertainty market sentiment, and oil uncertainty market sentiment. The data ranges from October, 2004 to July, 2023. The following table

describes in detail the data attributes, sample periods, and data sources of each variable used in the study.

Table 1: Data description

Variable	Sample Period	Description	Data Sources
WTI	2004M10-2023M7	West Texas Intermediate (WTI) crude oil spot price	Federal Reserve Economic Data
Gold	2004M10-2023M7	Gold prices	Investing.com
S&P500	2004M10-2023M7	The Standard and Poor's 500	Investing.com
USDI	2004M10-2023M7	US dollar index	Investing.com
EPU_News_Based_Policy_Uncert_Index	2004M10-2023M7	US Uncertainty Market Sentiment	policyuncertainty.com
OPU	2004M10-2023M7	Oil Uncertainty Market Sentiment	policyuncertainty.com
Brent	2004M10-2023M7	Brent crude oil spot prices	Federal Reserve Economic Data
Gold	2004M10-2023M7	Gold prices	Investing.com
ASX50	2004M10-2023M7	The ASX 50	Investing.com
AUDI	2004M10-2023M7	Euro index	Investing.com
EPU_News_Based_Policy_Uncert_Index	2004M10-2023M7	AU Uncertainty Market Sentiment	policyuncertainty.com
OPU	2004M10-2023M7	Oil Uncertainty Market Sentiment	policyuncertainty.com

To ensure consistency and ease of comparison, all variables were scaled using min-max scaling into the range of [0, 1]. This transformation was done to retain the relative differences while making them suitable for comparison analysis.

5. Expected Results

The study is expected to uncover dependencies and insights in the data that we collected and extrapolate conclusions or advice for future exploitation of the results according to eventual price fluctuation forecasts or political decisions.

This study aims to provide a greater understanding of factors affecting oil prices. In addition, this study also aims to provide actionable insights, particularly during extreme market conditions.

Given there is limited research on the Australian market, we are also focusing on factors that are coming from the Australian market to provide regional insights on factors which affect oil prices.

Sentiment analysis and its impact is an area that is under-researched, particularly on its relationship with oil prices. Thus, a potential new contribution from this research is to explore the impact of market sentiment on oil prices.

6. Conclusion

In conclusion, the present proposal seeks to improve the knowledge of extreme interdependencies across financial markets by implementing Quantile Time-Varying Parameter Vector Autoregression (TVP-VAR) model. Different to traditional mean-based measures, the above method provides a more detailed examination of the dynamic linkage effects across extreme volatility quantiles than would have been the case with related markets, such as stock index, foreign exchange, oil, and gold. Incorporating the rising bullish or bearish mood in analyses of connections in the market behavior provides new levels of detail regarding how various developments, especially those associated with momentous changes in global, decentral and pandemics, disseminate to different market states. Such analysis will help understand the elasticity of network connections with regard to both the peak and nadir of market turmoil. The study is expected to provide pertinent information to investors, financial participants, and also policymakers regarding the always-changing risks and the relational aspects that could arise in times of severe situations such as market disruption. This find will be helpful in designing strategies for managing risks and also overall economic reforms in place that can effectively work in normal as well as in disturbed and turmoil situations, if possible. This will ultimately contribute to more stable financial systems and resilient market structures in the face of growing uncertainty.

7. Source Codes

Please view the current state of the Source Code and preliminary findings of descriptive analysis and EDA results for the US Data from the following link: [Source Code US](#)

Or:

<https://colab.research.google.com/drive/1lg6j-ht5tcc0BKkAhG02X2N2Lcn9Xt0p?usp=sharing>

For the Australian market data from the following link: [Source Code AU](#)

Or:

<https://colab.research.google.com/drive/1nwd1MvGXoj3paHWzCXt0iiqFdyJlzCyu?usp=sharing>

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